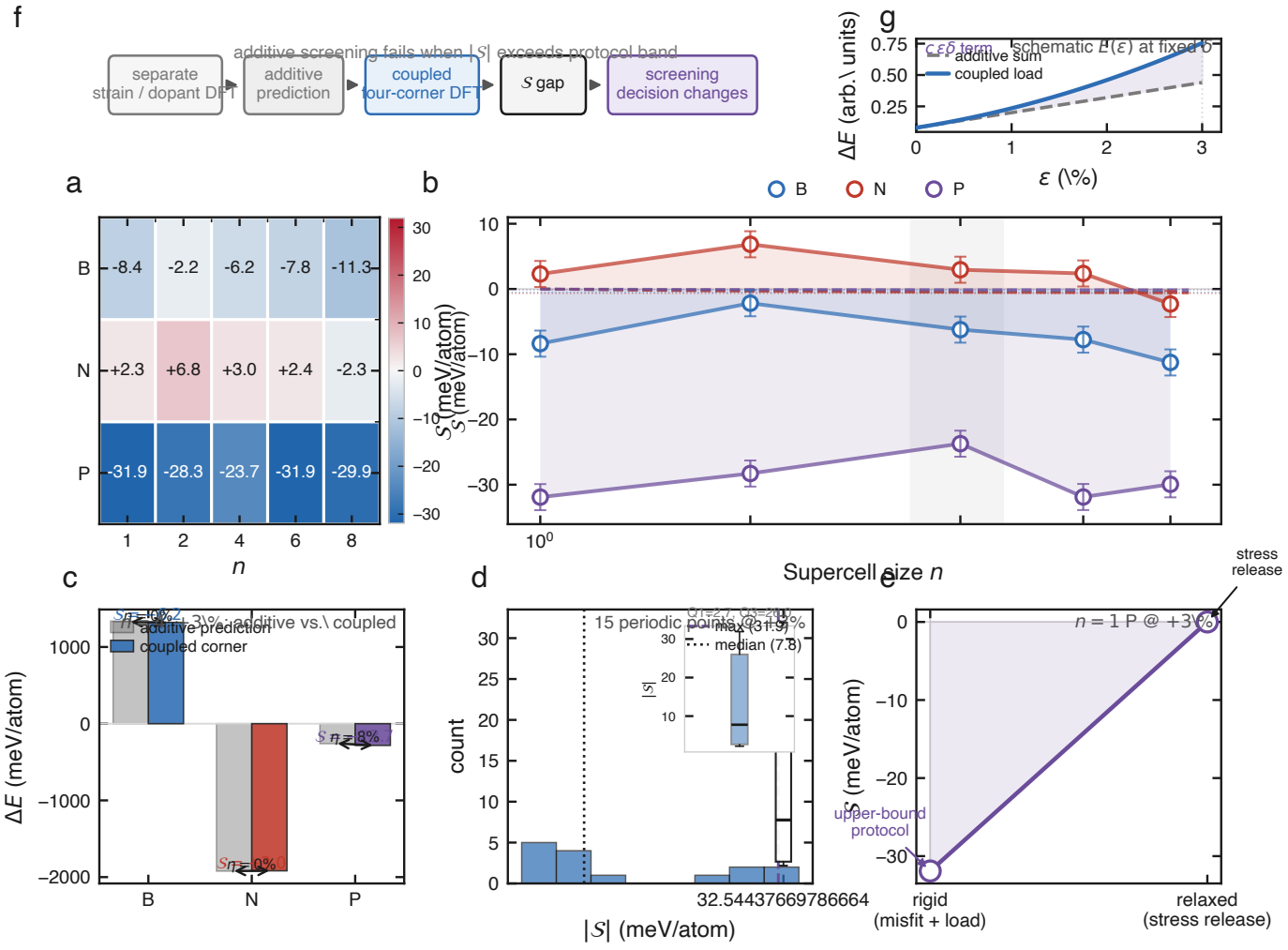
f
separate strain / dopant DFT
additive prediction
additive screening fails when |S| exceeds protocol band
coupled four-corner DFT
S gap
screening decision changes
g
schematic E(ε) at fixed δ
cεδ term
additive sum
coupled load
ΔE (arb.\ units)
ε (\%)
a
B
N
P
-8.4
-2.2
-6.2
-7.8
-11.3
+2.3
+6.8
+3.0
+2.4
-2.3
-31.9
-28.3
-23.7
-31.9
-29.9
n
b
S (meV/atom)
Supercell size n
c
ΔE (meV/atom)
additive prediction
coupled corner
additive vs.\ coupled
d
15 periodic points
max (31.9)
median (7.8)
count
|S| (meV/atom)
32.54437669786664
e
stress release
n = 1 P @ +3%
upper-bound protocol
rigid (misfit + load)
relaxed (stress release)